LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Hydraulic system, general design requirements, Typical layout of a basic hydraulic system and its brief description
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Hydraulic system, general design requirements, Typical layout of a basic hydraulic system and its brief description
5. Specific Objectives	17.1.2.1 Trainee will be able to describe typical hydraulic system and construction of its components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of hydraulic schematic diagrams; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a hydraulic system in aircraft?
(b) Name two advantages of hydraulic systems over mechanical linkages.
(c) What is the standard pressure range for typical aircraft hydraulic systems?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• General design requirements for hydraulic systems • Selection of working fluid and pressure limits • Safety factors and redundancy requirements • Material compatibility and corrosion resistance • System efficiency and heat dissipation needs	L & Ds	PPT, IB	15 min
(b)	• Typical layout of a basic hydraulic system • Function of reservoirs and fluid conditioning units • Operation of pumps and pressure control valves • Distribution networks and actuator components • Return lines and filtration systems	L & Ds	PPT, PDF	20 min
(c)	• Brief description of major system components • Construction details of pumps and motors • Valve types and their operational mechanisms • Actuator construction and force generation • System integration and interconnection logic	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of the reservoir in maintaining system pressure stability.
(b) Describe how a pressure relief valve protects the hydraulic system from over-pressurization.
(c) Identify the path of hydraulic fluid from the pump to the actuator and back.

12. Home Assignment and Follow Up	Prepare a brief note on the importance of clean pipelines and protective ointment when handling hydraulic oils, linking this to the next lesson on oil specifications.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Specification of hydraulic oil, need for clean pipelines and use of protective ointment while handling oils.	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min